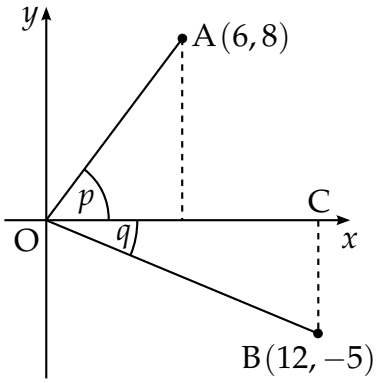
Higher 2000 - Paper 1 Solutions

[SQA] 1. On the coordinate diagram shown, A is the point $(6, 8)$ and B is the point $(12, -5)$. Angle $\text{AOC} = p$ and angle $\text{COB} = q$.

Find the exact value of $\sin(p + q)$. **4**



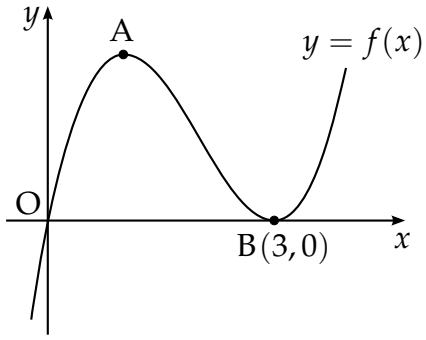
Part	Marks	Level	Calc.	Content	Answer	U2 OC3
	4	C	NC	T9	$\frac{63}{65}$	2000 P1 Q1

•1 ss: know to use trig expansion •2 pd: process missing sides •3 ic: interpret data •4 pd: process	•1 $\sin p \cos q + \cos p \sin q$ •2 10 **and** 13 •3 $\frac{8}{10} \cdot \frac{12}{13} + \frac{6}{10} \cdot \frac{5}{13}$ •4 $\frac{126}{130}$

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[SQA] 2. A sketch of the graph of $y = f(x)$ where $f(x) = x^3 - 6x^2 + 9x$ is shown below.

The graph has a maximum at A and a minimum at $B(3, 0)$.



(*a*) Find the coordinates of the turning point at A. **4**

(*b*) Hence sketch the graph of $y = g(x)$ where $g(x) = f(x + 2) + 4$.

Indicate the coordinates of the turning points. There is no need to calculate the coordinates of the points of intersection with the axes. **2**

(*c*) Write down the range of values of k for which $g(x) = k$ has 3 real roots. **1**

Part	Marks	Level	Calc.	Content	Answer	U1 OC3
(*a*)	4	C	NC	C8	$A(1, 4)$	2000 P1 Q2
(*b*)	2	C	NC	A3	sketch (translate 4 up, 2 left)	
(*c*)	1	A/B	NC	A2	$4 < k < 8$	

- $\bullet^1$ ss: know to differentiate
- $\bullet^2$ pd: differentiate correctly
- $\bullet^3$ ss: know gradient $= 0$
- $\bullet^4$ pd: process
- $\bullet^5$ ic: interpret transformation
- $\bullet^6$ ic: interpret transformation
- $\bullet^7$ ic: interpret sketch

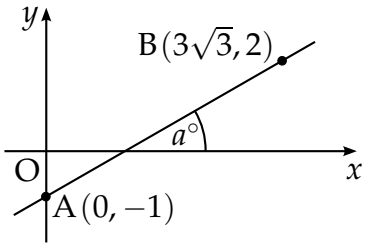
- $\bullet^1$ $\frac{dy}{dx} = \ldots$
- $\bullet^2$ $\frac{dy}{dx} = 3x^2 - 12x + 9$
- $\bullet^3$ $3x^2 - 12x + 9 = 0$
- $\bullet^4$ $A = (1, 4)$

translate $f(x)$ 4 units up, 2 units left

- $\bullet^5$ sketch with coord. of $A'(-1, 8)$
- $\bullet^6$ sketch with coord. of $B'(1, 4)$
- $\bullet^7$ $4 < k < 8$ (accept $4 \leq k \leq 8$)

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[SQA] 3. Find the size of the angle a° that the line joining the points $A(0,-1)$ and $B(3\sqrt{3},2)$ makes with the positive direction of the x-axis.



3

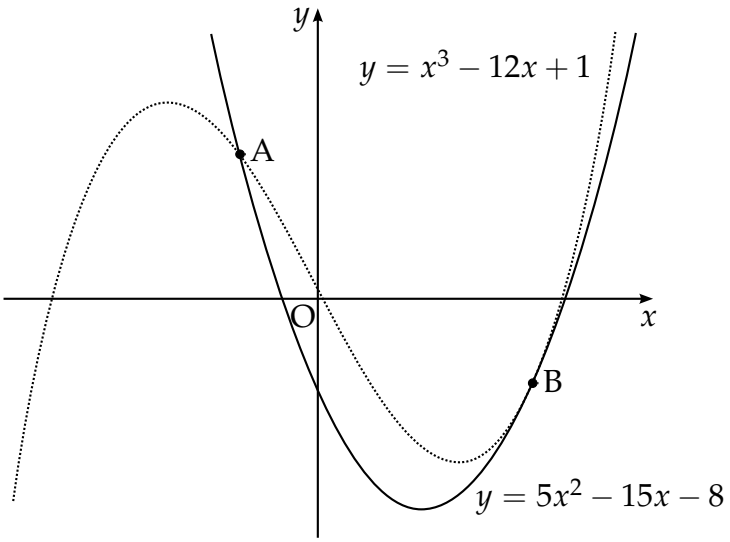
Part	Marks	Level	Calc.	Content	Answer	U1 OC1
	3	C	NC	G2	30	2000 P1 Q3

•1 ss: know how to find gradient or equ. •2 pd: process •3 ic: interpret exact value	•1 $\frac{2-(-1)}{3\sqrt{3}-0}$ •2 $\tan a =$ gradient *stated or implied by* •3 •3 $a = 30$

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[SQA] 4. The diagram shows a sketch of the graphs of $y = 5x^2 - 15x - 8$ and $y = x^3 - 12x + 1$.

The two curves intersect at A and touch at B, i.e. at B the curves have a common tangent.



(*a*) (i) Find the x-coordinates of the point of the curves where the gradients are equal. **4**

(ii) By considering the corresponding y-coordinates, or otherwise, distinguish geometrically between the two cases found in part (i). **1**

(*b*) The point A is $(-1, 12)$ and B is $(3, -8)$.

Find the area enclosed between the two curves. **5**

Part	Marks	Level	Calc.	Content	Answer	U2 OC2
(*ai*)	4	C	NC	C4	$x = \frac{1}{3}$ and $x = 3$	2000 P1 Q4
(*aii*)	1	C	NC	CGD	parallel and coincident	
(*b*)	5	C	NC	C17	$21\frac{1}{3}$	

$\bullet^1$ ss: know to diff. and equate $\bullet^2$ pd: differentiate $\bullet^3$ pd: form equation $\bullet^4$ ic: interpret solution $\bullet^5$ ic: interpret diagram $\bullet^6$ ss: know how to find area between curves $\bullet^7$ ic: interpret limits $\bullet^8$ pd: form integral $\bullet^9$ pd: process integration $\bullet^{10}$ pd: process limits	$\bullet^1$ find derivatives and equate $\bullet^2$ $3x^2 - 12$ **and** $10x - 15$ $\bullet^3$ $3x^2 - 10x + 3 = 0$ $\bullet^4$ $x = 3$, $x = \frac{1}{3}$ $\bullet^5$ tangents at $x = \frac{1}{3}$ are parallel, at $x = 3$ coincident $\bullet^6$ $\int(\text{cubic} - \text{parabola})$ **or** $\int(\text{cubic}) - \int(\text{parabola})$ $\bullet^7$ $\int_{-1}^{3} \cdots dx$ $\bullet^8$ $\int(x^3 - 5x^2 + 3x + 9)dx$ or equiv. $\bullet^9$ $\left[\frac{1}{4}x^4 - \frac{5}{3}x^3 + \frac{3}{2}x^2 + 9x\right]_{-1}^{3}$ or equiv. $\bullet^{10}$ $21\frac{1}{3}$

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[SQA] 5. Two sequences are generated by the recurrence relations $u_{n+1} = au_n + 10$ and $v_{n+1} = a^2v_n + 16$.

The two sequences approach the same limit as $n \to \infty$.

Determine the value of a and evaluate the limit. **5**

Part	Marks	Level	Calc.	Content	Answer	U1 OC4
	4	C	NC	A13	$a = \frac{3}{5}, L = 25$	2000 P1 Q5
	1	A/B	NC	A12		

$\bullet^1$ ss: know how to find limit $\bullet^2$ pd: process $\bullet^3$ pd: process $\bullet^4$ ic: interpret coeff. of u_n $\bullet^5$ pd: process	$\bullet^1$ $L = aL + 10$ **or** $L = a^2L + 16$ **or** $L = \frac{b}{1-a}$ $\bullet^2$ $L = \frac{10}{1-a}$ **or** $L = \frac{16}{1-a^2}$ $\bullet^3$ $\frac{10}{1-a}$ **or** $\frac{16}{1-a^2}$ $\bullet^4$ $10a^2 - 16a + 6 = 0$ $\bullet^5$ $a = \frac{3}{5}$ **and** $L = 25$

[SQA] 6. For what range of values of k does the equation $x^2 + y^2 + 4kx - 2ky - k - 2 = 0$ represent a circle? **5**

Part	Marks	Level	Calc.	Content	Answer	U2 OC4
	5	A	NC	G9, A17	for all k	2000 P1 Q6

$\bullet^1$ ss: know to examine radius $\bullet^2$ pd: process $\bullet^3$ pd: process $\bullet^4$ ic: interpret quadratic inequation $\bullet^5$ ic: interpret quadratic inequation	$\bullet^1$ $g = 2k$, $f = -k$, $c = -k - 2$ *stated or implied by* $\bullet^2$ $\bullet^2$ $r^2 = 5k^2 + k + 2$ $\bullet^3$ (real $r \Rightarrow$) $5k^2 + k + 2 > 0$ (*accept* $\geq$) $\bullet^4$ use discr. **or** complete sq. **or** diff. $\bullet^5$ true for all k

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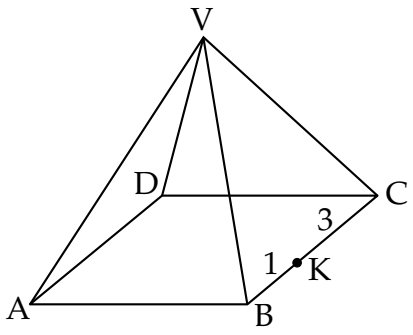
[SQA] 7. VABCD is a pyramid with a rectangular base ABCD.

Relative to some appropriate axes,

$\overrightarrow{VA}$ represents $-7\boldsymbol{i} - 13\boldsymbol{j} - 11\boldsymbol{k}$

$\overrightarrow{AB}$ represents $6\boldsymbol{i} + 6\boldsymbol{j} - 6\boldsymbol{k}$

$\overrightarrow{AD}$ represents $8\boldsymbol{i} - 4\boldsymbol{j} + 4\boldsymbol{k}$.



K divides BC in the ratio 1 : 3.

Find $\overrightarrow{VK}$ in component form. **3**

Part	Marks	Level	Calc.	Content	Answer	U3 OC1
	3	C	CN	G25, G21, G20	$\begin{pmatrix} 1 \\ -8 \\ -16 \end{pmatrix}$	2000 P1 Q7

$\bullet^1$ ss: recognise crucial aspect $\bullet^2$ ic: interpret ratio $\bullet^3$ pd: process components	$\bullet^1$ $\overrightarrow{VK} = \overrightarrow{VA} + \overrightarrow{AB} + \overrightarrow{BK}$ **or** $\overrightarrow{VK} = \overrightarrow{VB} + \overrightarrow{BK}$ $\bullet^2$ $\overrightarrow{BK} = \frac{1}{4}\overrightarrow{BC}$ **or** $\frac{1}{4}\overrightarrow{AD}$ **or** $\begin{pmatrix} 2 \\ -1 \\ 1 \end{pmatrix}$ **or** $\begin{pmatrix} -1 \\ -7 \\ -17 \end{pmatrix}$ $\bullet^3$ $\overrightarrow{VK} = \begin{pmatrix} 1 \\ -8 \\ -16 \end{pmatrix}$

[SQA] 8. The graph of $y = f(x)$ passes through the point $\left(\frac{\pi}{9}, 1\right)$.

If $f'(x) = \sin(3x)$ express y in terms of x. **4**

Part	Marks	Level	Calc.	Content	Answer	U3 OC2
	4	A/B	NC	C18, C23	$y = -\frac{1}{3}\cos(3x) + \frac{7}{6}$	2000 P1 Q8

$\bullet^1$ ss: know to integrate $\bullet^2$ pd: integrate $\bullet^3$ ic: interpret $(\frac{\pi}{9}, 1)$ $\bullet^4$ pd: process	$\bullet^1$ $y = \int \sin(3x)\,dx$ *stated or implied by* $\bullet^2$ $\bullet^2$ $-\frac{1}{3}\cos(3x)$ $\bullet^3$ $1 = -\frac{1}{3}\cos(\frac{3\pi}{9}) + c$ or equiv. $\bullet^4$ $c = \frac{7}{6}$

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[SQA] 9. Evaluate $\log_5 2 + \log_5 50 - \log_5 4$. **3**

Part	Marks	Level	Calc.	Content	Answer	U3 OC3
	2	C	NC	A28	2	2000 P1 Q9
	1	A/B	NC	A28		

•[1] pd: use $\log_a x + \log_a y = \log_a xy$ •[2] pd: use $\log_a x - \log_a y = \log_a \frac{x}{y}$ •[3] pd: use $\log_a a = 1$	•[1] $\log_5 100 - \log_5 4$ •[2] $\log_5 25$ •[3] 2

[SQA] 10. Find the maximum value of $\cos x - \sin x$ and the value of x for which it occurs in the interval $0 \leq x \leq 2\pi$. **6**

Part	Marks	Level	Calc.	Content	Answer	U3 OC4
	6	A/B	CN	T14	max value $\sqrt{2}$ when $x = \frac{7\pi}{4}$	2000 P1 Q10

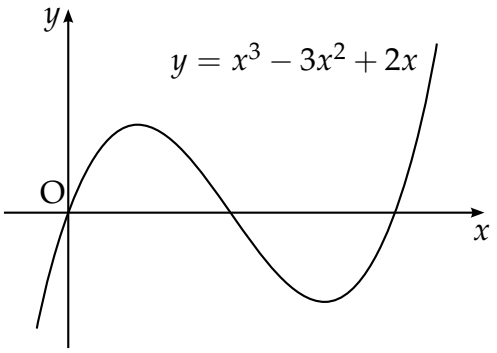
•[1] ss: use e.g. $k\cos(x+a)$ •[2] ic: expand chosen rule •[3] pd: compare coefficients •[4] pd: process •[5] pd: process •[6] ic: interpret trig expression	•[1] e.g. use $k\cos(x+a)$ •[2] $k\cos x\cos a - k\sin x\sin a$ •[3] $k\cos a = 1$ and $k\sin a = 1$ •[4] $k = \sqrt{2}$ •[5] $\tan a = 1$, $a = \frac{\pi}{4}$ ($45°$ *is bad form*) •[6] max. value $= \sqrt{2}$ when $x = \frac{7\pi}{4}$ (*do not accept* $45°$)

[END OF QUESTIONS]

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Higher 2000 - Paper 2 Solutions

[SQA] 1. The diagram shows a sketch of the graph of $y = x^3 - 3x^2 + 2x$.



(*a*) Find the equation of the tangent to this curve at the point where $x = 1$. **5**

(*b*) The tangent at the point $(2, 0)$ has equation $y = 2x - 4$. Find the coordinates of the point where this tangent meets the curve again. **5**

Part	Marks	Level	Calc.	Content	Answer	U2 OC1
(*a*)	5	C	CN	C5	$x + y = 1$	2000 P2 Q1
(*b*)	5	C	CN	A23, A22, A21	$(-1, -6)$	

$\bullet^1$ ss: know to differentiate
$\bullet^2$ pd: differentiate correctly
$\bullet^3$ ss: know that gradient $= f'(1)$
$\bullet^4$ ss: know that y-coord $= f(1)$
$\bullet^5$ ic: state equ. of line

$\bullet^6$ ss: equate equations
$\bullet^7$ pd: arrange in standard form
$\bullet^8$ ss: know how to solve cubic
$\bullet^9$ pd: process
$\bullet^{10}$ ic: interpret

$\bullet^1$ $y' = \ldots$
$\bullet^2$ $3x^2 - 6x + 2$
$\bullet^3$ $y'(1) = -1$
$\bullet^4$ $y(1) = 0$
$\bullet^5$ $y - 0 = -1(x - 1)$

$\bullet^6$ $2x - 4 = x^3 - 3x^2 + 2x$
$\bullet^7$ $x^3 - 3x^2 + 4 = 0$
$\bullet^8$

$\ldots$	1	-3	0	4
		$\ldots$	$\ldots$	$\ldots$
	$\ldots$	$\ldots$	$\ldots$	$\ldots$

$\bullet^9$ identify $x = -1$ from working
$\bullet^{10}$ $(-1, -6)$

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[SQA] 2. (*a*) Find the equation of AB, the perpendicular bisector of the line joing the points $P(-3,1)$ and $Q(1,9)$. **4**

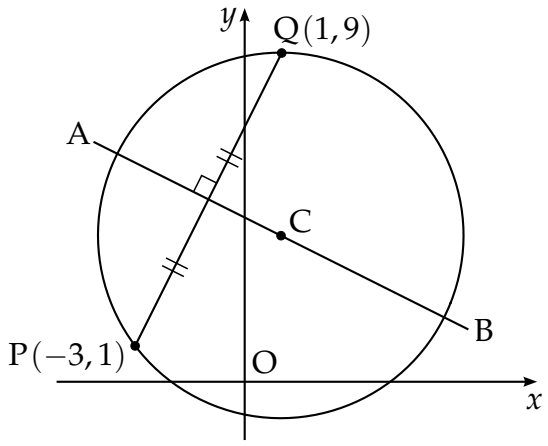
(*b*) C is the centre of a circle passing through P and Q. Given that QC is parallel to the y-axis, determine the equation of the circle. **3**

(*c*) The tangents at P and Q intersect at T.

Write down

(i) the equation of the tangent at Q

(ii) the coordinates of T. **2**



Part	Marks	Level	Calc.	Content	Answer	U2 OC4
(*a*)	4	C	CN	G7	$x+2y=9$	2000 P2 Q2
(*b*)	3	C	CN	G10	$(x-1)^2+(y-4)^2=25$	
(*c*)	2	C	CN	G11, G8	(i) $y=9$, (ii) $T(-9,9)$	

$\bullet^1$ ss: know to use midpoint $\bullet^2$ pd: process gradient of PQ $\bullet^3$ ss: know how to find perp. gradient $\bullet^4$ ic: state equ. of line $\bullet^5$ ic: interpret "parallel to y-axis" $\bullet^6$ pd: process radius $\bullet^7$ ic: state equ. of circle $\bullet^8$ ic: interpret diagram $\bullet^9$ ss: know to use equ. of AB	$\bullet^1$ midpoint $=(-1,5)$ $\bullet^2$ $m_{PQ}=\frac{9-1}{1-(-1)}$ $\bullet^3$ $m_\perp=-\frac{1}{2}$ $\bullet^4$ $y-5=-\frac{1}{2}(x-(-1))$ $\bullet^5$ $y_C=4$ *stated or implied by* $\bullet^7$ $\bullet^6$ radius $=5$ or equiv. *stated or implied by* $\bullet^7$ $\bullet^7$ $(x-1)^2+(y-4)^2=25$ $\bullet^8$ $y=9$ $\bullet^9$ T$=(-9,9)$

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[SQA] 3. $f(x) = 3 - x$ and $g(x) = \dfrac{3}{x}, x \neq 0$.

(*a*) Find $p(x)$ where $p(x) = f(g(x))$. **2**

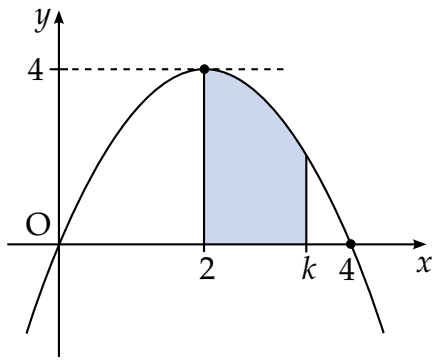
(*b*) If $q(x) = \dfrac{3}{3-x}, x \neq 3$, find $p(q(x))$ in its simplest form. **3**

Part	Marks	Level	Calc.	Content	Answer	U1 OC2
(*a*)	2	C	CN	A4	$3 - \frac{3}{x}$	2000 P2 Q3
(*b*)	2	C	CN	A4	x	
(*b*)	1	A/B	CN	A4		

$\bullet^1$ ic: interpret composite func. $\bullet^2$ pd: process	$\bullet^1$ $f\left(\frac{3}{x}\right)$ *stated or implied by* $\bullet^2$ $\bullet^2$ $3 - \frac{3}{x}$
$\bullet^3$ ic: interpret composite func. $\bullet^4$ pd: process $\bullet^5$ pd: process	$\bullet^3$ $p\left(\frac{3}{3-x}\right)$ *stated or implied by* $\bullet^4$ $\bullet^4$ $3 - \frac{3}{\frac{3}{3-x}}$ $\bullet^5$ x

[SQA] 4. The parabola shown crosses the x-axis at $(0,0)$ and $(4,0)$, and has a maximum at $(2,4)$.

The shaded area is bounded by the parabola, the x-axis and the lines $x = 2$ and $x = k$.



(*a*) Find the equation of the parabola. **2**

(*b*) Hence show that the shaded area, A, is given by

$A = -\frac{1}{3}k^3 + 2k^2 - \frac{16}{3}$. **3**

Part	Marks	Level	Calc.	Content	Answer	U2 OC2
(*a*)	2	C	CN	A19	$y = 4x - x^2$	2000 P2 Q4
(*b*)	3	C	CN	C16	proof	

$\bullet^1$ ic: state standard form $\bullet^2$ pd: process for x^2 coeff.	$\bullet^1$ $ax(x-4)$ $\bullet^2$ $a = -1$
$\bullet^3$ ss: know to integrate $\bullet^4$ pd: integrate correctly $\bullet^5$ pd: process limits and complete proof	$\bullet^3$ $\int_2^k$(function from (a)) $\bullet^4$ $-\frac{1}{3}x^3 + 2x^2$ $\bullet^5$ $-\frac{1}{3}k^3 + 2k^2 - \left(-\frac{8}{3} + 8\right)$

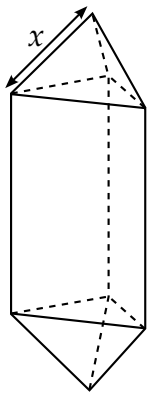
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[SQA] 5. Solve the equation $3\cos 2x^\circ + \cos x^\circ = -1$ in the interval $0 \leq x \leq 360$. **5**

Part	Marks	Level	Calc.	Content	Answer	U2 OC3
	5	A/B	CR	T10	60, 131·8, 228·2, 300	2000 P2 Q5

•1 ss: know to use $\cos 2x = 2\cos^2 x - 1$ •2 pd: process •3 ss: know to/and factorise quadratic •4 pd: process •5 pd: process	•1 $3(2\cos^2 x^\circ - 1)$ •2 $6\cos^2 x^\circ + \cos x^\circ - 2 = 0$ •3 $(2\cos x^\circ - 1)(3\cos x^\circ + 2)$ •4 $\cos x^\circ = \frac{1}{2}$, $x = 60, 30$ •5 $\cos x^\circ = -\frac{2}{3}$, $x = 132, 228$

[SQA] 6. A goldsmith has built up a solid which consists of a triangular prism of fixed volume with a regular tetrahedron at each end.



The surface area, A, of the solid is given by

$$A(x) = \frac{3\sqrt{3}}{2}\left(x^2 + \frac{16}{x}\right)$$

where x is the length of each edge of the tetrahedron.

Find the value of x which the goldsmith should use to minimise the amount of gold plating required to cover the solid. **6**

Part	Marks	Level	Calc.	Content	Answer	U1 OC3
	6	A/B	CN	C11	$x = 2$	2000 P2 Q6

•1 ss: know to differentiate •2 pd: process •3 ss: know to set $f'(x) = 0$ •4 pd: deal with x^{-2} •5 pd: process •6 ic: check for minimum	•1 $A'(x) = \ldots$ •2 $\frac{3\sqrt{3}}{2}(2x - 16x^{-2})$ **or** $3\sqrt{3}x - 24\sqrt{3}x^{-2}$ •3 $A'(x) = 0$ •4 $-\frac{16}{x^2}$ **or** $-\frac{24\sqrt{3}}{x^2}$ •5 $x = 2$ •6 $\begin{array}{c\|ccc} x & 2^- & 2 & 2^+ \\ \hline A'(x) & -ve & 0 & +ve \end{array}$ so $x = 2$ is min.

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[SQA] 7. For what value of t are the vectors $\boldsymbol{u} = \begin{pmatrix} t \\ -2 \\ 3 \end{pmatrix}$ and $\boldsymbol{v} = \begin{pmatrix} 2 \\ 10 \\ t \end{pmatrix}$ perpendicular? **2**

Part	Marks	Level	Calc.	Content	Answer	U3 OC1
	2	C	CN	G27	$t = 4$	2000 P2 Q7

•1 ss: know to use scalar product •2 ic: interpret scalar product	•1 $\boldsymbol{u}.\boldsymbol{v} = 2t - 20 + 3t$ •2 $\boldsymbol{u}.\boldsymbol{v} = 0 \Rightarrow t = 4$

[SQA] 8. Given that $f(x) = (5x - 4)^{\frac{1}{2}}$, evaluate $f'(4)$. **3**

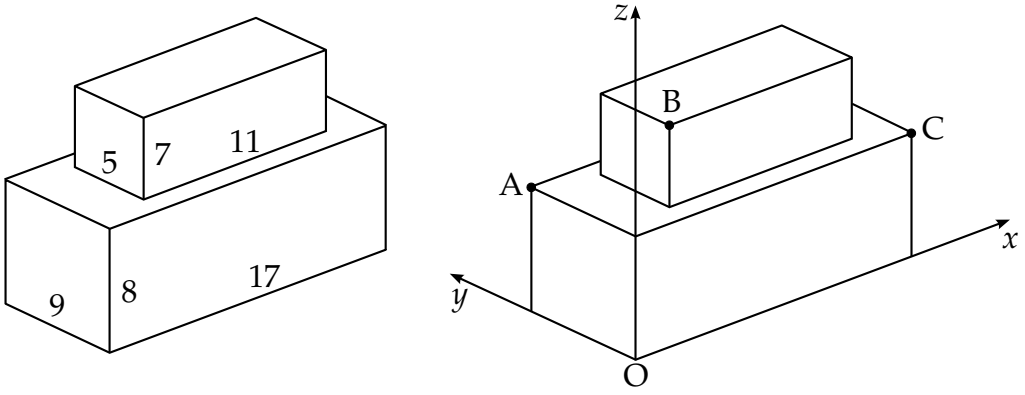
Part	Marks	Level	Calc.	Content	Answer	U3 OC2
	1	C	CN	C21	$\frac{5}{8}$	2000 P2 Q8
	2	A/B	CN	C21		

•1 pd: differentiate power •2 pd: differentiate 2nd function •3 pd: evaluate $f'(x)$	•1 $\frac{1}{2}(5x - 4)^{-\frac{1}{2}}$ •2 $\times 5$ •3 $f'(4) = \frac{5}{8}$

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[SQA] 9. A cuboid measuring 11 cm by 5 cm by 7 cm is placed centrally on top of another cuboid measuring 17 cm by 9 cm by 8 cm.

Coordinates axes are taken as shown.



(*a*) The point A has coordinates $(0, 9, 8)$ and C has coordinates $(17, 0, 8)$.
Write down the coordinates of B. **1**

(*b*) Calculate the size of angle ABC. **6**

Part	Marks	Level	Calc.	Content	Answer	U3 OC1
(*a*)	1	C	CN	G22	B$(3, 2, 15)$	2000 P2 Q9
(*b*)	6	C	CR	G28	$92{\cdot}5°$	

- $\bullet^1$ ic: interpret 3-d representation
- $\bullet^2$ ss: know to use scalar product
- $\bullet^3$ pd: process vectors
- $\bullet^4$ pd: process vectors
- $\bullet^5$ pd: process lengths
- $\bullet^6$ pd: process scalar product
- $\bullet^7$ pd: evaluate scalar product

- $\bullet^1$ B$= (3, 2, 15)$ *treat* $\begin{pmatrix} 3 \\ 2 \\ 15 \end{pmatrix}$ *as bad form*
- $\bullet^2$ $\cos \text{A}\widehat{\text{B}}\text{C} = \frac{\overrightarrow{\text{BA}}.\overrightarrow{\text{BC}}}{|\overrightarrow{\text{BA}}||\overrightarrow{\text{BC}}|}$
- $\bullet^3$ $\overrightarrow{\text{BA}} = \begin{pmatrix} -3 \\ 7 \\ -7 \end{pmatrix}$
- $\bullet^4$ $\overrightarrow{\text{BC}} = \begin{pmatrix} 14 \\ -2 \\ -7 \end{pmatrix}$
- $\bullet^5$ $|\overrightarrow{\text{BA}}| = \sqrt{107}$, $|\overrightarrow{\text{BC}}| = \sqrt{249}$
- $\bullet^6$ $\overrightarrow{\text{BA}}.\overrightarrow{\text{BC}} = -7$
- $\bullet^7$ $\text{A}\widehat{\text{B}}\text{C} = 92{\cdot}5°$

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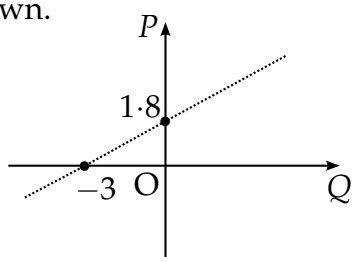
[SQA] 10. Find $\int \frac{1}{(7-3x)^2}\,dx$. **2**

Part	Marks	Level	Calc.	Content	Answer	U3 OC2
	2	A/B	CN	C22, C14	$\frac{1}{3(7-3x)}+c$	2000 P2 Q10

•1 pd: integrate function •2 pd: deal with function of function	•1 $\frac{1}{-1}(7-3x)^{-1}$ •2 $\times\frac{1}{-3}$

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[SQA] 11. The results of an experiment give rise to the graph shown.



(*a*) Write down the equation of the line in terms of P and Q. **2**

It is given that $P = \log_e p$ and $Q = \log_e q$.

(*b*) Show that p and q satisfy a relationship of the form $p = aq^b$, stating the values of a and b. **4**

Part	Marks	Level	Calc.	Content	Answer	U3 OC3
(*a*)	2	A/B	CR	G3	$P = 0{\cdot}6Q + 1{\cdot}8$	2000 P2 Q11
(*b*)	4	A/B	CR	A33	$a = 6{\cdot}05$, $b = 0{\cdot}6$	

- $\bullet^1$ ic: interpret gradient
- $\bullet^2$ ic: state equ. of line
- $\bullet^3$ ic: interpret straight line
- $\bullet^4$ ss: know how to deal with x of $x \log y$
- $\bullet^5$ ss: know how to express number as log
- $\bullet^6$ ic: interpret sum of two logs

- $\bullet^1$ $m = \frac{1{\cdot}8}{3} = 0{\cdot}6$
- $\bullet^2$ $P = 0{\cdot}6Q + 1{\cdot}8$

Method 1

- $\bullet^3$ $\log_e p = 0{\cdot}6 \log_e q + 1{\cdot}8$
- $\bullet^4$ $\log_e q^{0{\cdot}6}$
- $\bullet^5$ $\log_e 6{\cdot}05$
- $\bullet^6$ $p = 6{\cdot}05q^{0{\cdot}6}$

Method 2

$\ln p = \ln aq^b$

- $\bullet^3$ $\ln p = \ln a + b \ln q$
- $\bullet^4$ $\ln p = 0{\cdot}6 \ln q + 1{\cdot}8$ *stated or implied by* $\bullet^5$ *or* $\bullet^6$
- $\bullet^5$ $\ln a = 1{\cdot}8$
- $\bullet^6$ $a = 6{\cdot}05$, $b = 0{\cdot}6$

[END OF QUESTIONS]

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